A. Learning report

A short report covering:

```
1. Why ViT?
2. How image patchification works
3. Patch embeddings
4. Positional embeddings
5. CLS token
6. ViT Transformer architecture
7. CLS vs patch tokens
8. What DINOv2 is
9. What DINOv2 outputs
10.How DINOv2 will be used in this project
```

B. Colab notebook

Something like:

```
Week1_DINOv2_DESI.ipynb

├── Environment setup
├── Load DESI dataset
├── Dataset exploration
├── Visualize g/r/i/z
├── Load pretrained DINOv2
├── Preprocess image
├── Run inference
├── Extract CLS token
├── Extract patch tokens
├── Inspect tensor dimensions
├── Visualize patch grid
└── Initial embedding exploration
```

C. GitHub

```
Week-1/
├── README.md
├── Week1_DINOv2_DESI.ipynb
```

```
└── report/
    └── Week1_Report.pdf
```

```
ViT paper
   ↓
Understand ViT
   ↓
DESI dataset
   ↓
Understand astronomical images
   ↓
Load pretrained DINOv2
   ↓
Run inference
   ↓
Extract CLS + patch tokens
   ↓
Understand what these representations mean
```